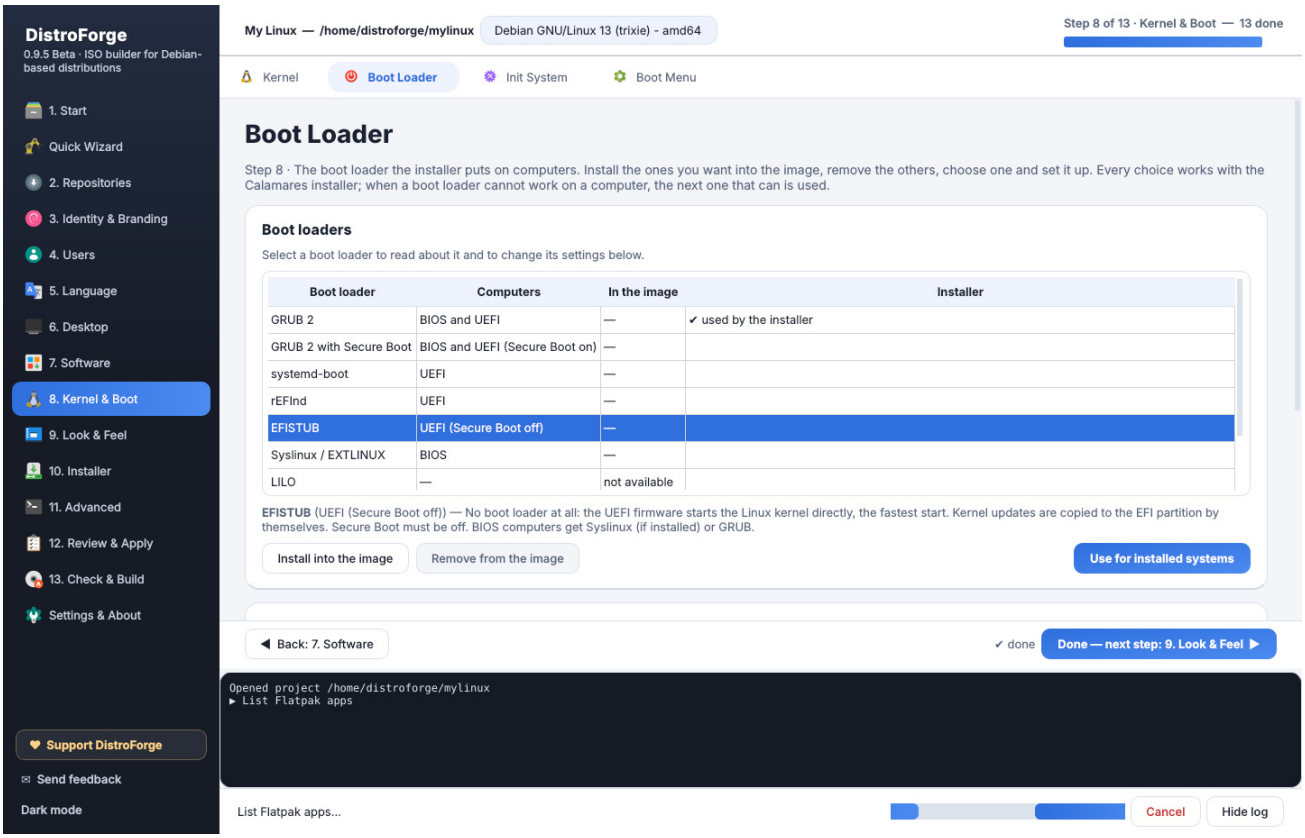
DistroForge

User Guide · 0.9.5 Beta

Build your own Debian-based Linux distribution as a live ISO, step by step.



Hugo Moniz do Rego · GPL-3.0-or-later · https://github.com/hugomonizdorego/Eduka-Customizer

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1. What is DistroForge?

DistroForge is free and open source software (GPL-3.0-or-later) to build your own Linux distribution from Debian or a Debian derivative. The result is a live ISO that boots from a USB stick or DVD (BIOS and UEFI) and installs with the Calamares installer.

You can start from:

- a live ISO of Debian or a Debian derivative (for example LMDE, MX Linux, antiX, Kali Linux, Parrot OS, Deepin, SparkyLinux, BunsenLabs, Q4OS)
- a new Debian base system (stable, testing or sid) made with mmdebstrap/debootstrap
- a copy of this computer (snapshot)

Devuan (Debian without systemd) and its derivatives, such as Refracta, Exe GNU/Linux and Gnuinos, work too: DistroForge then writes Devuan's repositories (deb.devuan.org) and can bootstrap a new Devuan base.

Ubuntu-based ISOs (Ubuntu, Linux Mint's Ubuntu edition, Pop!_OS, elementary OS, Zorin OS) are not supported: DistroForge is made for Debian-based distributions. DistroForge itself installs on Debian, Debian derivatives and Ubuntu.

The work is split into 13 steps. The next step opens when the one before is done, so you always know what comes next. Experts can open every step in Settings.

- 1 Start · 2 Repositories · 3 Identity & Branding · 4 Users · 5 Language · 6 Desktop
- 7 Software · 8 Kernel & Boot · 9 Look & Feel · 10 Installer · 11 Advanced
- 12 Review & Apply · 13 Check & Build

2. Installing and starting

Install the .deb package (it pulls in every tool it needs):

```
sudo apt install ./distroforge_0.9.5~beta_all.deb
```

Start it from the application menu (**DistroForge**) or a terminal:

```
distroforge            # the window (asks for the admin password)
sudo distroforge --help  # command line
```

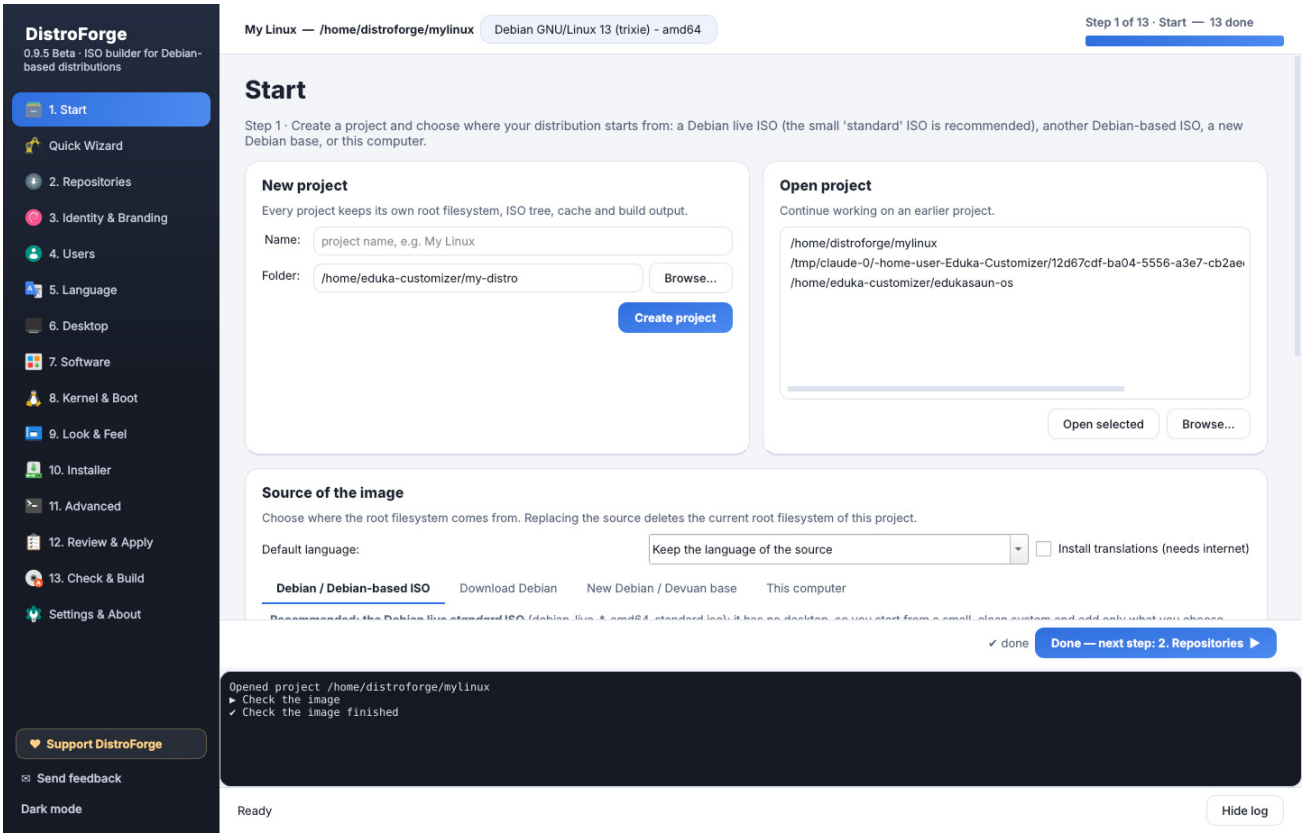
The old command `eduka-customizer` still works and opens DistroForge. Old settings in /etc/eduka-customizer are read automatically.

Needs: Debian 12 or newer (or a derivative), 4 GB RAM, free disk space of three times the ISO size.

3. Step 1 · Start: create a project

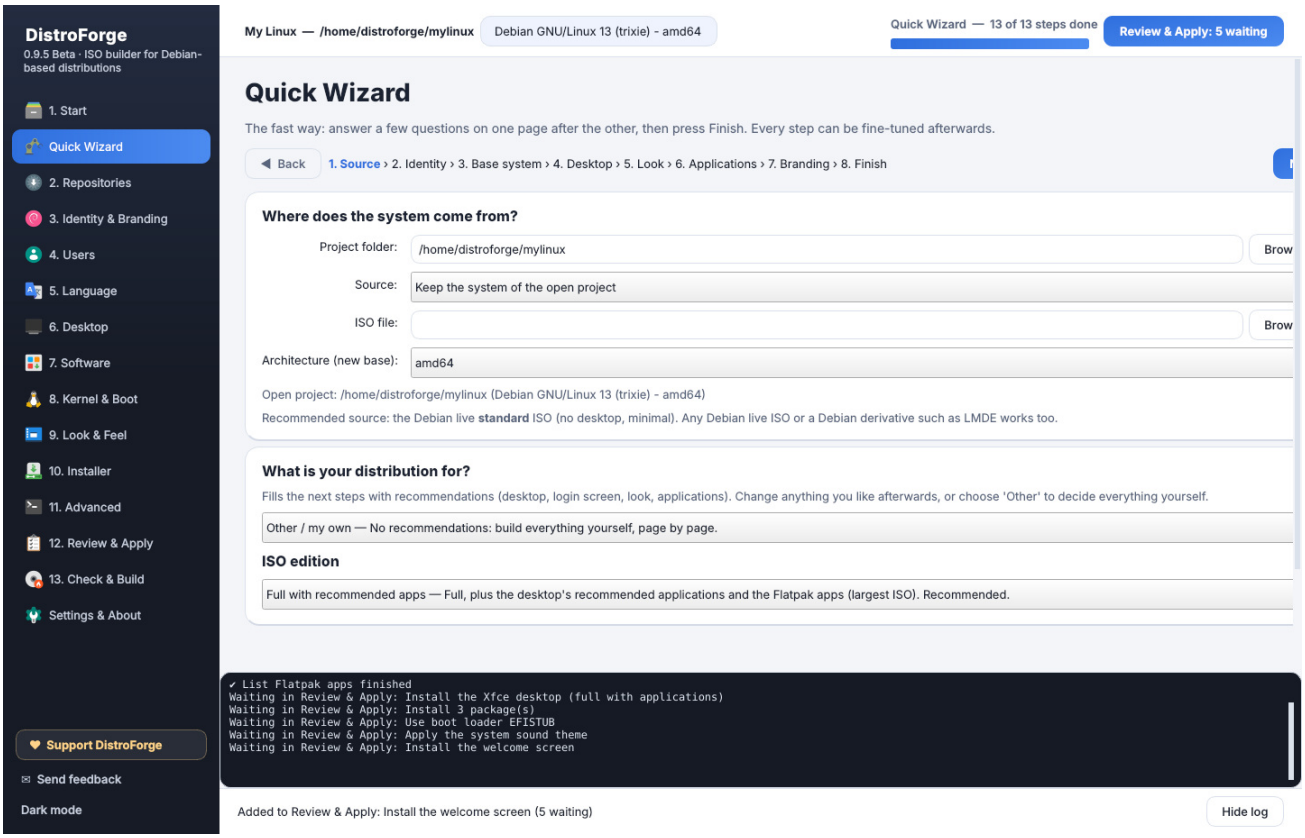
Choose where your distribution comes from: open an ISO, download an official Debian live ISO, bootstrap a new Debian base, or copy this computer. Each project is one folder with the system, the settings and the built ISO, each in its own sub-folder (see 'The project folder').

Purpose asks what the distribution is for (home, office, school, gaming, server, ...) and the ISO edition (minimal, standard, full). Use or ignore its recommendations.



Step 1: create or open a project

The **Quick Wizard** asks everything important on 8 pages and prepares it all at once.

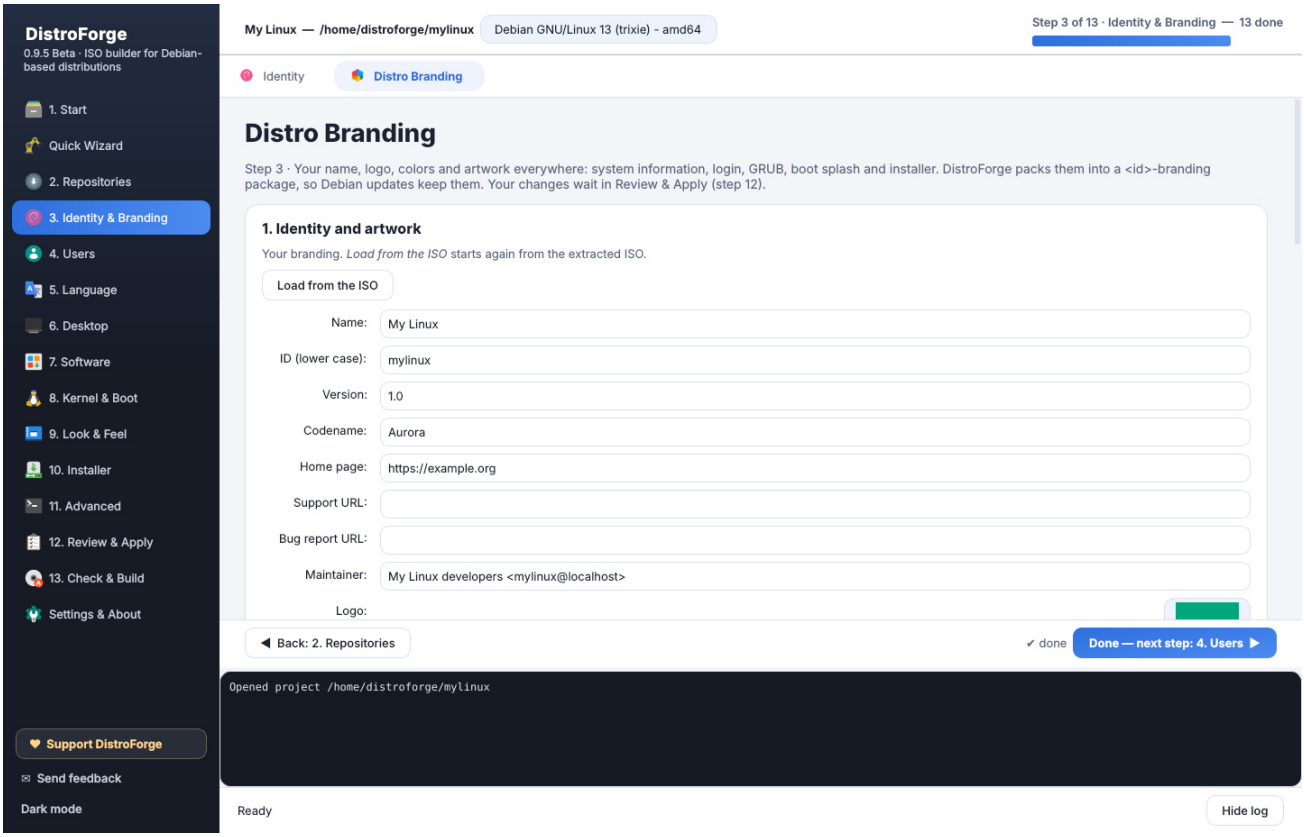


Quick Wizard

4. Steps 2 to 5 · Repositories, Identity, Users, Language

Repositories: the Debian version (stable, testing, sid), the mirror, contrib/non-free-firmware and other repositories.

Identity & Branding: name, version, code name, web site, logo and wallpaper. The values start from what the ISO has. DistroForge builds a branding package that replaces os-release, lsb-release, issue, logos, the GRUB menu and the installer branding. Use a name and logo of your own: names such as 'Debian ...' or 'Ubuntu ...' are trademarks, and the check warns about them.

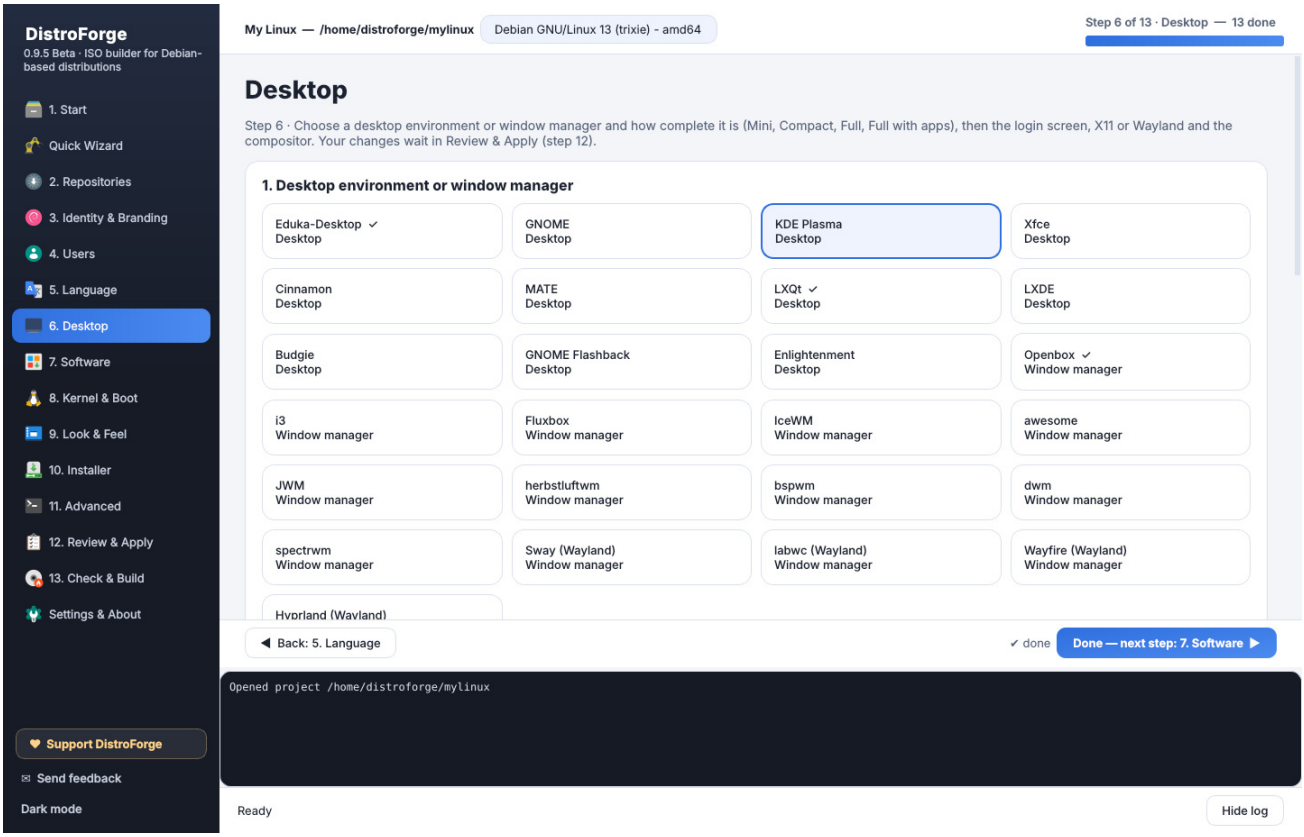


Identity & Branding

Users: the live user (default `live`, full name `Live`), with or without a password, automatic login and more accounts. **Language**: default language, keyboard, time zone (default Asia/Dili) and language packs, also for the installer and the boot menu.

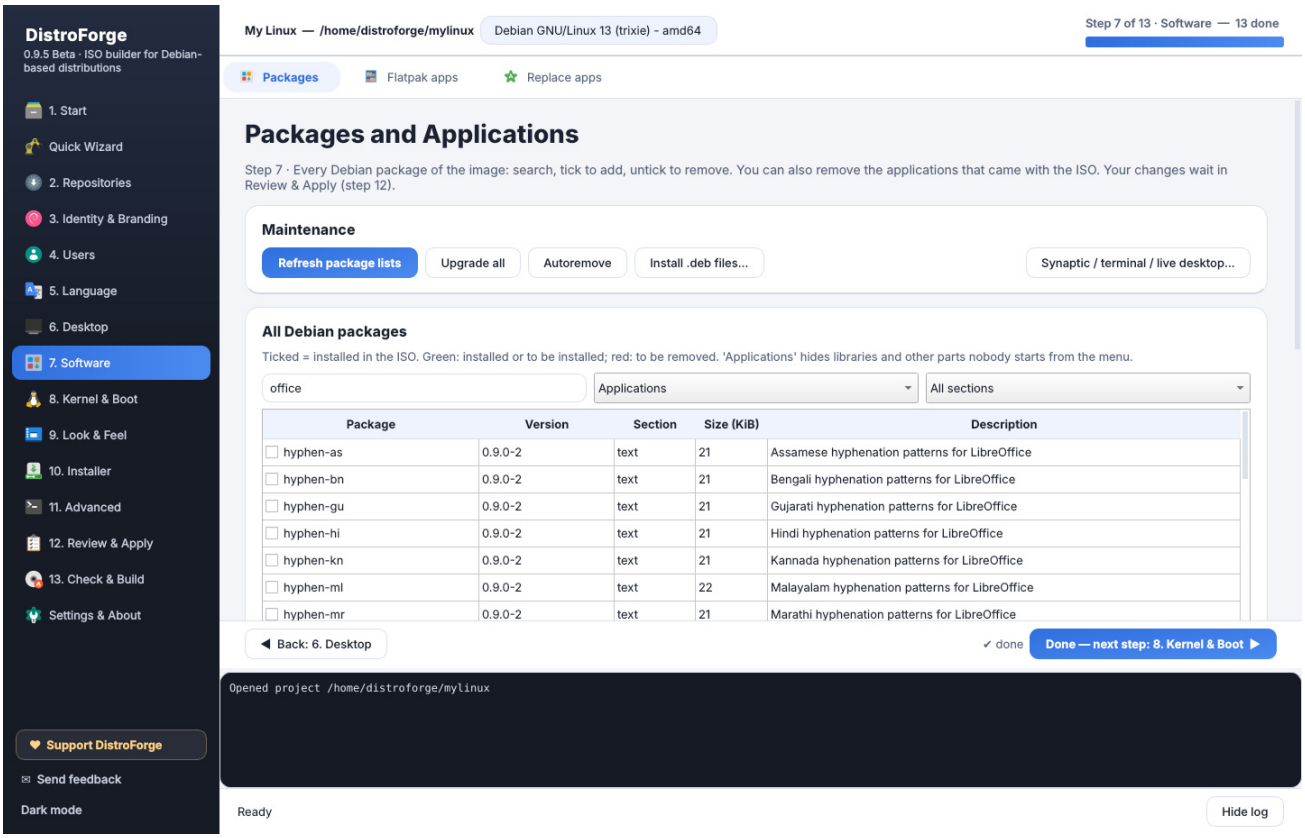
5. Steps 6 and 7 · Desktop and Software

Choose a desktop (GNOME, KDE Plasma, Xfce, Cinnamon, MATE, LXQt, Budgie, ...) or a window manager and its edition under the names the desktop itself uses. The login screen, X11 or Wayland and the compositor follow the chosen desktop (marked 'recommended'); change them if you like. Eduka-Desktop is one of the desktops: its source and its defaults for new users show when it is chosen. One button adds everything to Review & Apply.



Desktops and editions

Software: search, install and remove Debian packages; add Flathub applications by category; and replace the default applications (browser, mail, editor, terminal, ...) for all users.


DistroForge
0.9.5 Beta · ISO builder for Debian-based distributions
1. Start
Quick Wizard
2. Repositories
3. Identity & Branding
4. Users
5. Language
6. Desktop
7. Software
8. Kernel & Boot
9. Look & Feel
10. Installer
11. Advanced
12. Review & Apply
13. Check & Build
Settings & About
Support DistroForge
Send feedback
Dark mode
My Linux — /home/distroforge/mylinux
Debian GNU/Linux 13 (trixie) - amd64
Step 7 of 13 · Software — 13 done
Packages
Flatpak apps
Replace apps
Packages and Applications
Step 7 · Every Debian package of the image: search, tick to add, untick to remove. You can also remove the applications that came with the ISO. Your changes wait in Review & Apply (step 12).
Maintenance
Refresh package lists
Upgrade all
Autoremove
Install .deb files...
Synaptic / terminal / live desktop...
All Debian packages
Ticked = installed in the ISO. Green: installed or to be installed; red: to be removed. 'Applications' hides libraries and other parts nobody starts from the menu.
office
Applications
All sections
Package
Version
Section
Size (KiB)
Description
hyphen-as 0.9.0-2 text 21 Assamese hyphenation patterns for LibreOffice
hyphen-bn 0.9.0-2 text 21 Bengali hyphenation patterns for LibreOffice
hyphen-gu 0.9.0-2 text 21 Gujarati hyphenation patterns for LibreOffice
hyphen-hi 0.9.0-2 text 21 Hindi hyphenation patterns for LibreOffice
hyphen-kn 0.9.0-2 text 21 Kannada hyphenation patterns for LibreOffice
hyphen-ml 0.9.0-2 text 22 Malayalam hyphenation patterns for LibreOffice
hyphen-mr 0.9.0-2 text 21 Marathi hyphenation patterns for LibreOffice
◀ Back: 6. Desktop
✔ done
Done — next step: 8. Kernel & Boot ▶
Opened project /home/distroforge/mylinux
Ready
Hide log

Packages

6. Step 8 · Kernel & Boot

Four tabs: **Kernel** (Debian, backports or third-party kernels, firmware, drivers), **Boot Loader** (what installed computers start with), **Init System** and **Boot Menu** (the menu of the ISO itself).

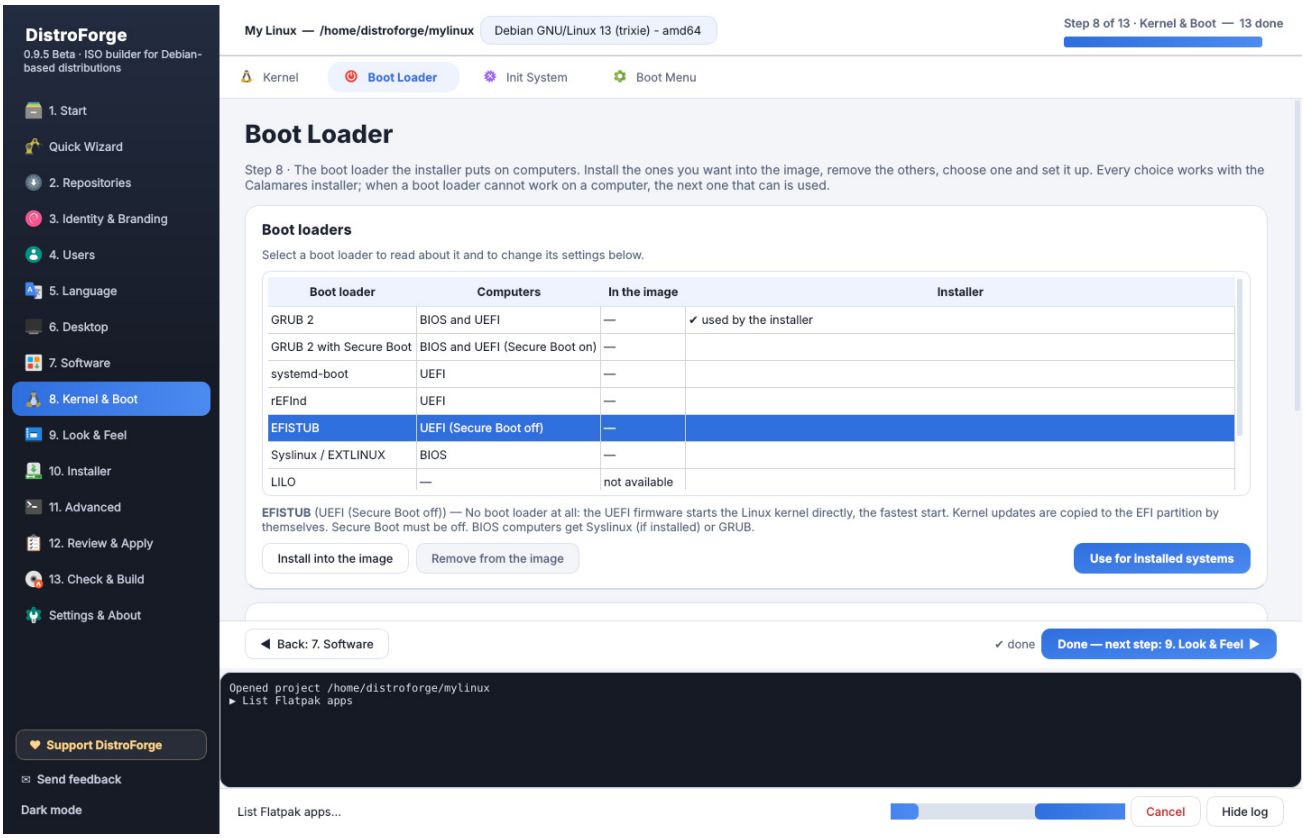
Kernel terminal. Copy the whole instructions from a kernel's web site and paste them into the terminal at once; 'sudo' and prompts are taken out, missing tools are installed and 'apt update' runs by itself. The kernels of the new repository are listed below it.

Init System. systemd (Debian's default), SysVinit (Devuan's default), OpenRC or runit. Switching also installs elogind for desktop logins, the right live-config backend and init scripts, and tells the Calamares installer. GNOME needs systemd. s6 and GNU Shepherd are explained but cannot start a Debian system.

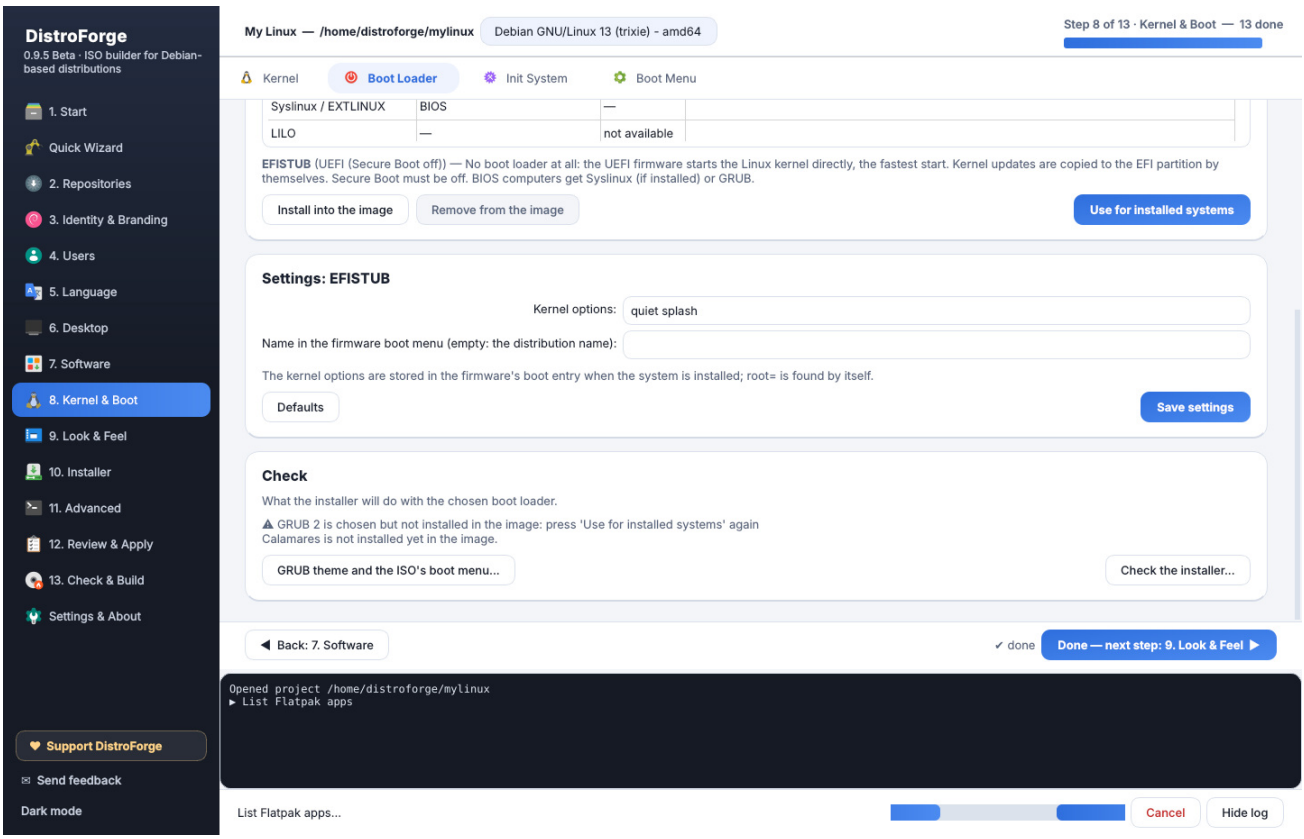
Boot Loader. Install the boot loaders you want into the image, remove the others, choose the one the installer puts on computers (**Use for installed systems**) and set it up below:

- **GRUB 2** — BIOS and UEFI, menus, themes, other operating systems. Recommended. Settings: timeout, kernel options, menu style, default entry, os-prober, recovery entries, resolution.
- **GRUB 2 with Secure Boot** — the same, signed by Debian and started by shim.
- **systemd-boot** — small and fast, UEFI. Settings: timeout, kernel options, default entry, editor, screen mode.
- **rEFInd** — graphical, finds every system by itself, UEFI (Calamares 3.3). Settings: timeout, kernel options, resolution, text mode, tools row.
- **EFISTUB** — no boot loader: the UEFI firmware starts the kernel directly. Secure Boot must be off. Settings: kernel options, name in the firmware menu.
- **Syslinux / EXTLINUX** — the small classic boot loader for BIOS computers (/boot on ext2/3/4, not encrypted). Settings: timeout, kernel options, menu title, graphical menu.

Every choice works with the Calamares installer. GRUB, systemd-boot and rEFInd are installed by Calamares itself; EFISTUB and Syslinux by an own installer step. When a boot loader cannot work on a computer (Syslinux on UEFI, EFISTUB on BIOS, an encrypted /boot) the next one that works is used, GRUB last, so the installation does not stop with an error. After kernel updates EFISTUB and Syslinux are kept up to date by themselves.

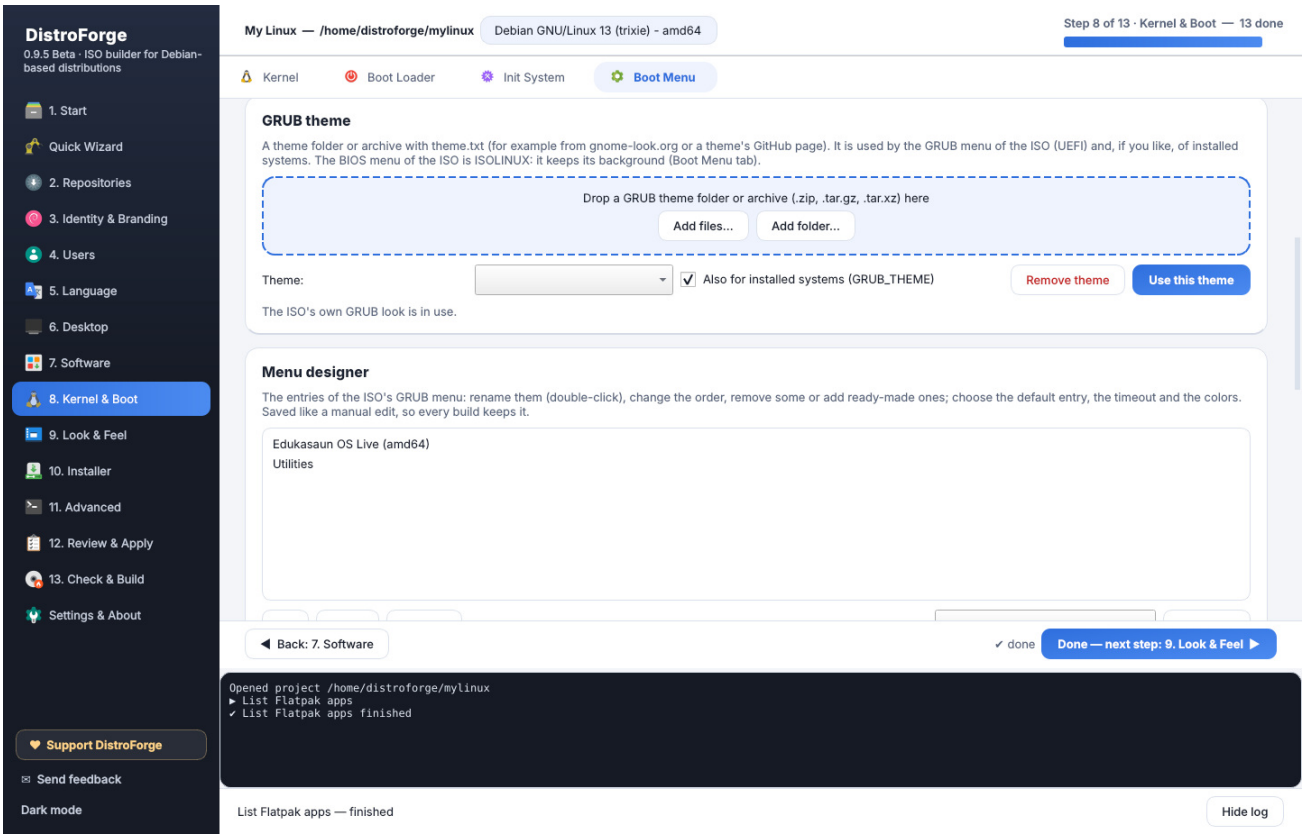


Boot Loader



The settings of the selected boot loader

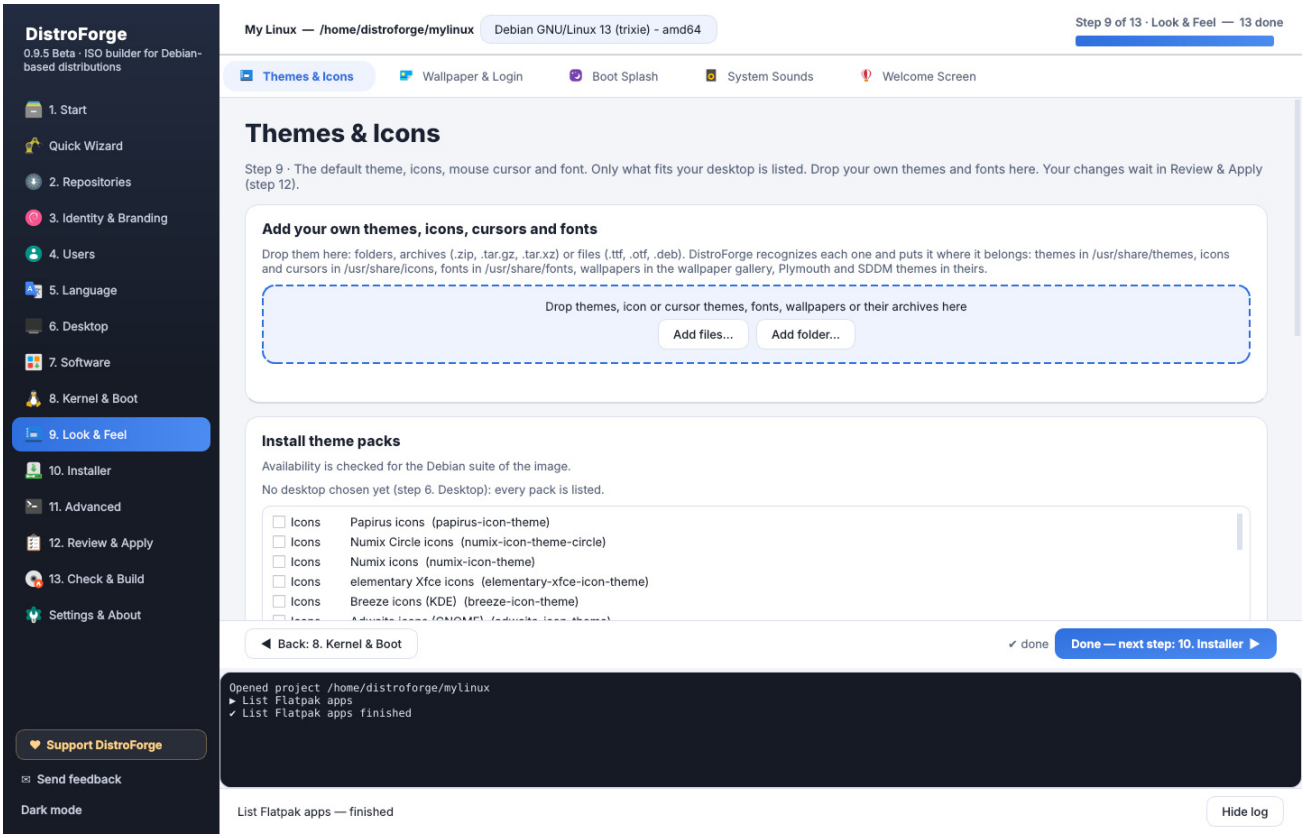
Boot Menu: title, timeout, kernel options and background of the ISO's menu, a third-party GRUB theme (checked first, refused when GRUB could not show it), the menu designer (rename, reorder, add or remove entries) and, for experts, the boot files themselves.



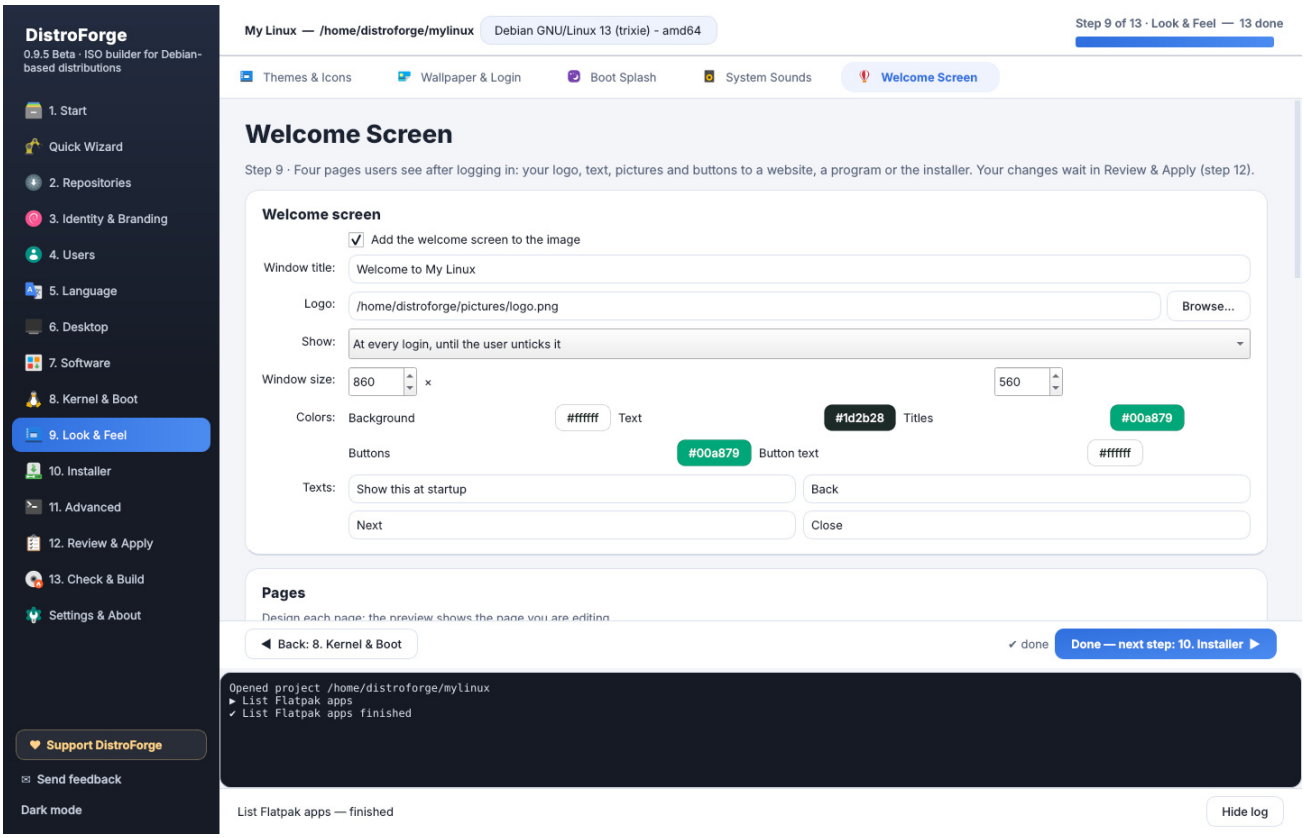
GRUB theme and menu designer

7. Step 9 · Look & Feel

Five tabs: **Themes & Icons** (only what fits the chosen desktop), **Wallpaper & Login**, **Plymouth** (boot animation), **System Sounds** (start-up, login, log out, errors, notifications) and **Welcome Screen** (four pages shown after the first login).



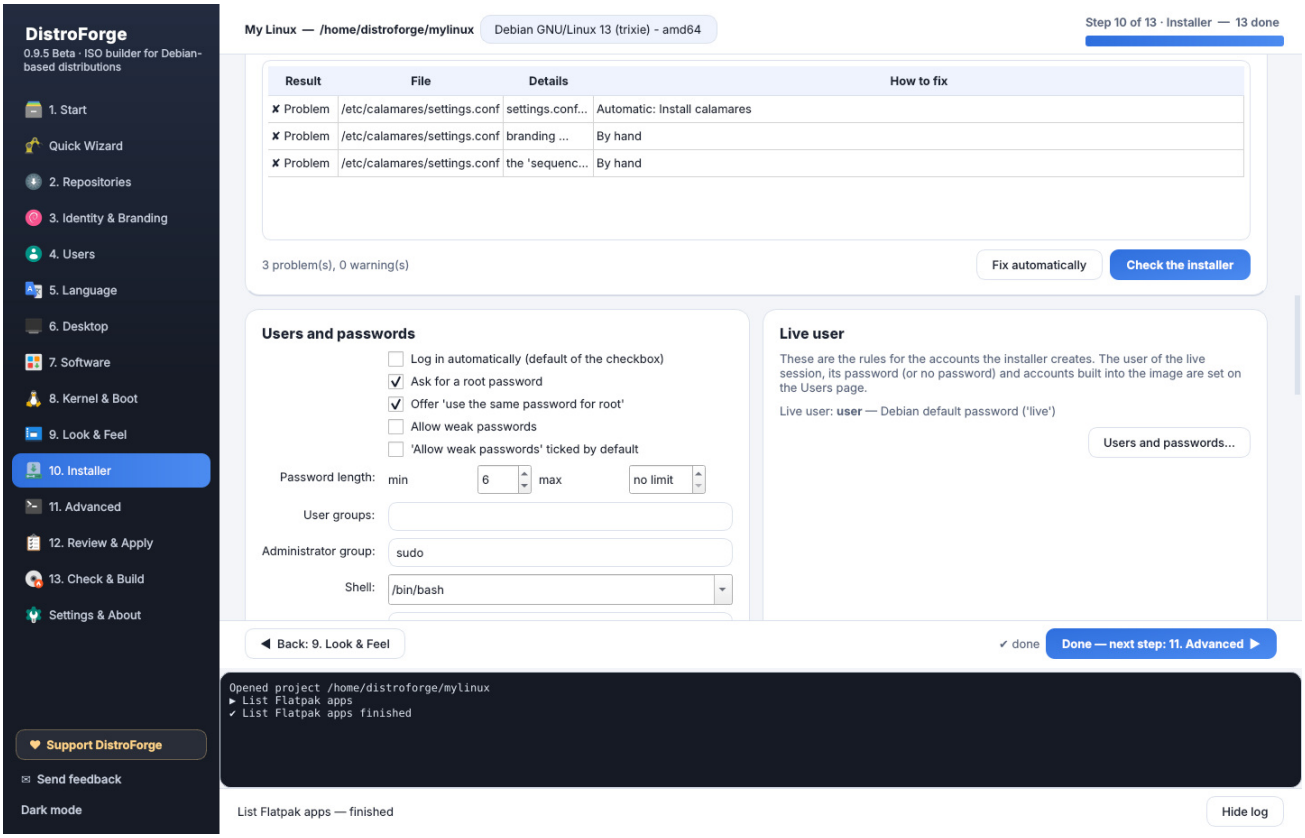
Themes and icons



Welcome screen

8. Steps 10 and 11 · Installer and Advanced

Installer: the Calamares installer — modules, order, slides, default partitioning and texts. **Check the installer** finds mistakes before the build; the **How to fix** column says whether a problem is fixed automatically.



Installer check

Advanced: a terminal inside the image, the live session in a window (Xephyr), hook scripts, JSON recipes and the package workshop to change files of installed Debian packages.

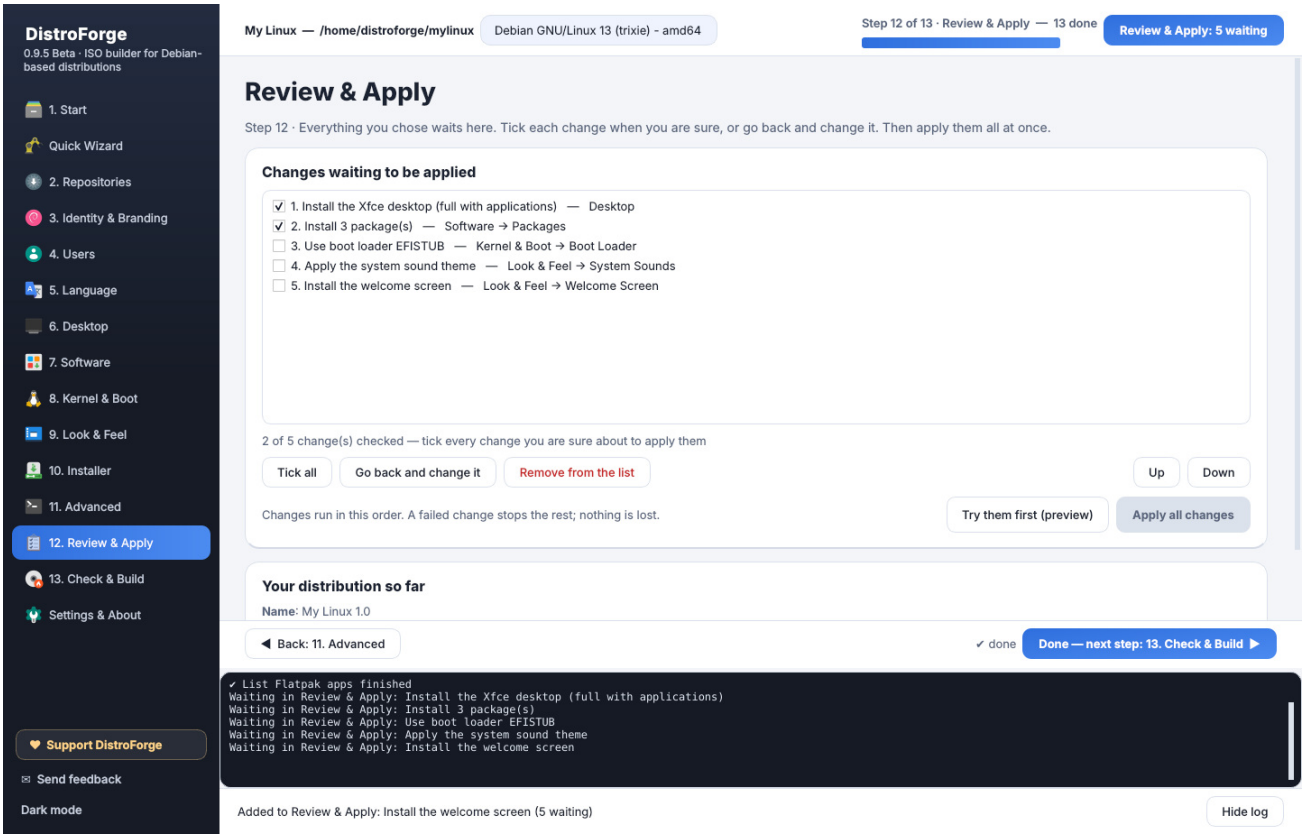
9. Step 12 · Review & Apply

Try them first (preview) applies the waiting changes to a temporary copy of the project. Start the live session (Advanced → Terminal & Live) to look at them, then **Keep them: apply for real** or **Throw the preview away** — the project is then exactly as before.

Changes from steps 2 to 11 **do not** change the image right away. They all wait in the Review & Apply list. The blue button at the top shows how many are waiting.

- Tick every change you are sure about.
- **Go back and change it** opens the step of that change.
- **Remove from the list** drops a change; **Up/Down** change the order.
- **Apply all changes** is enabled when everything is ticked. If one change fails, the ones applied before it leave the list and the rest keep waiting.

Experts can apply every change at once (Settings → Apply every change at once).



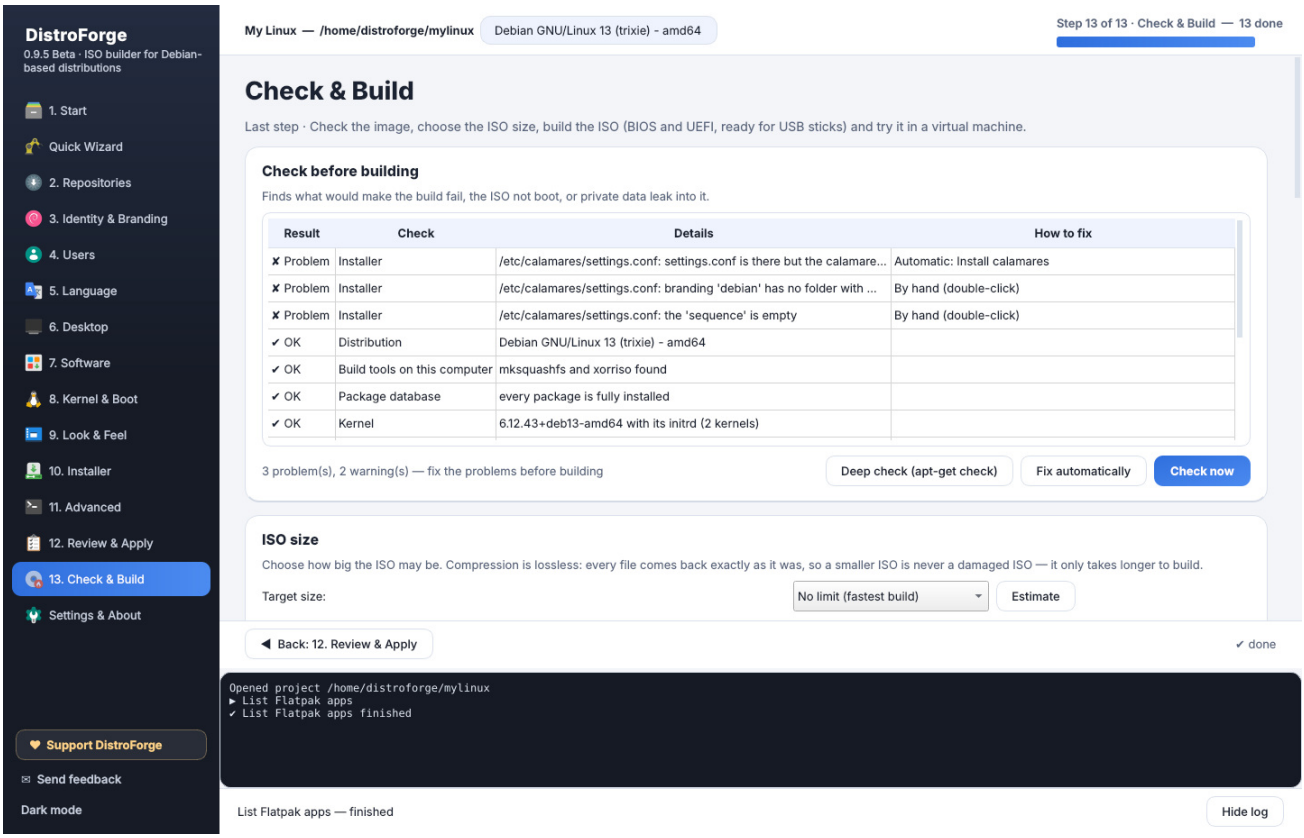
Review & Apply

10. Step 13 · Check & Build

Check now finds what would make the build fail, the ISO not boot, or private data leak into it. Every row says how to fix it:

- **Automatic: ...** — press **Fix automatically**; DistroForge fixes it and checks again.
- **By hand (double-click)** — double-click the row to open the right step.

When you press **Build ISO image** with problems left, DistroForge asks: **Fix automatically**, **I will fix it myself** or **Build anyway**.



Check & Build

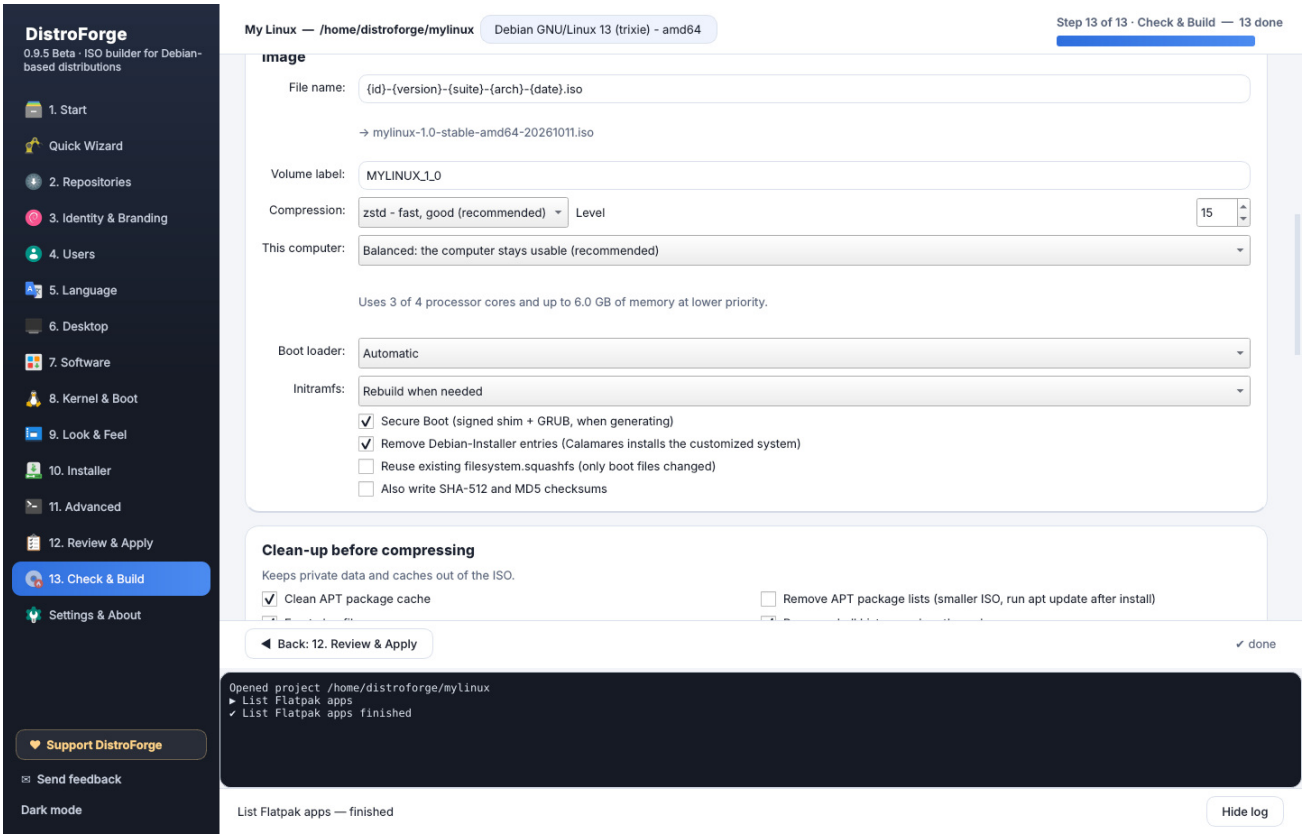
11. ISO size

Choose a target size: as small as possible, 100 MB, 300 MB, 500 MB, 700 MB (CD), 1 GB, 2 GB, 4.4 GB (DVD) or no limit. **Estimate** compresses a sample of the system and computes the size of every method.

Squashfs compression is **lossless**: every file comes back exactly as it was. A smaller ISO is never a damaged ISO, it only takes longer to build (xz is smallest and slowest, lz4 fastest).

This computer sets how much of the computer compressing may use: **Balanced** (one core stays free for you, lower priority, the default), **Full speed** (every core) or **Light** (half the cores at low priority). The cores and memory are taken from the computer itself.

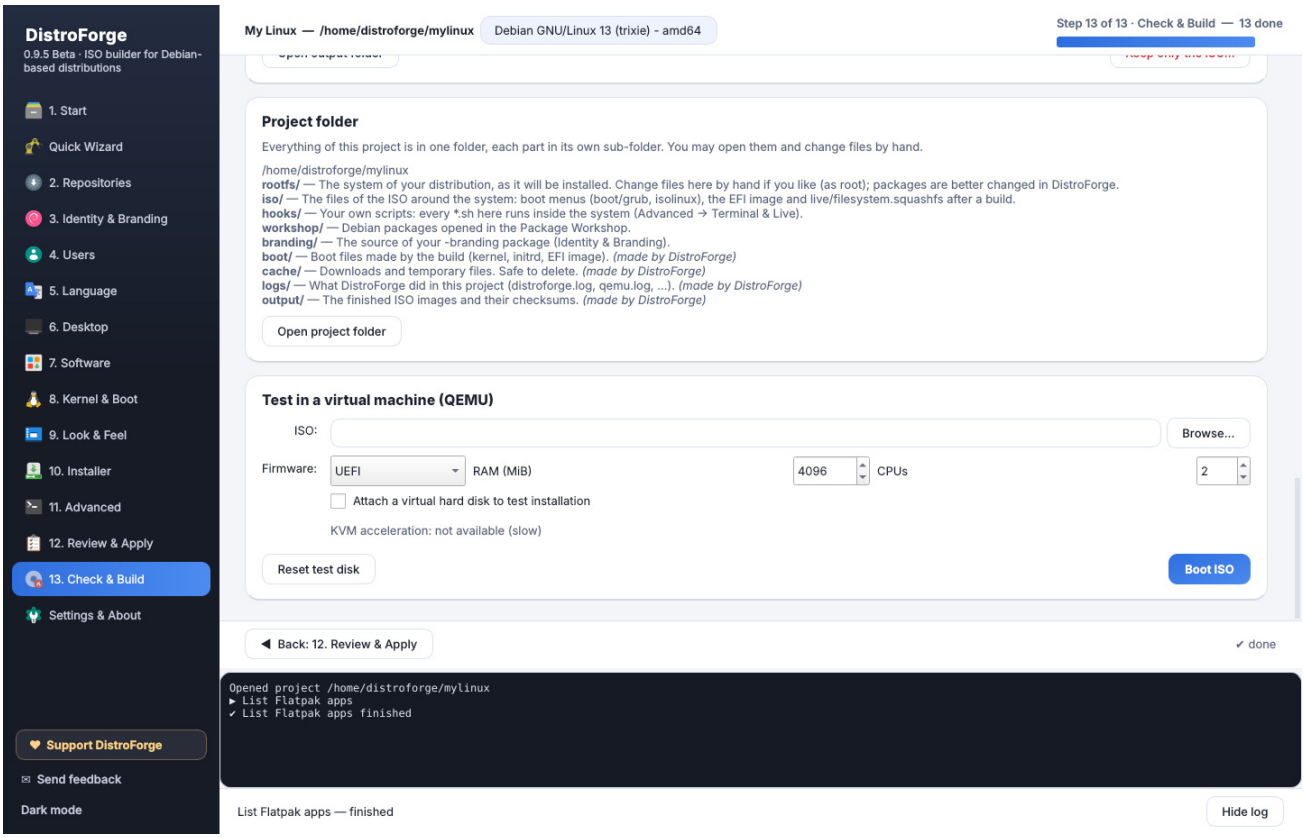
When the target is too small, DistroForge suggests safe size savers: documentation, manual pages, unused translations, APT lists and old kernels. License files always stay. Removing applications you do not need helps most.



Target size and build options

12. The project folder and keeping only the ISO

Everything of a project is in one folder, each part in its own sub-folder, so you can also open them and change files by hand: `rootfs/` (the system), `iso/` (boot menus and the files around the system), `hooks/` (your scripts), `workshop/`, `branding/`, and `boot/`, `cache/`, `logs/`, `output/` made by DistroForge. README.txt in the folder explains them.



Result and project folder

The build folders are big. When the ISO is finished and tested, **Keep only the ISO...** (or closing the window after a build) deletes everything of the project but the ISO and its checksum files, which move into the project folder. Nothing is deleted while anything is still mounted inside the project, and files of your own in the folder stay. The command line does the same with `distroforge clean --keep-iso`.

13. Testing and writing to USB

After the build, **Boot ISO** in the **Test in a virtual machine** card boots the ISO with QEMU (BIOS, UEFI or UEFI with Secure Boot). The ISO is hybrid: write it to a USB stick with GNOME Disks, balenaEtcher, Ventoy or:

```
sudo dd if=mylinux.iso of=/dev/sdX bs=4M status=progress oflag=sync
```

Replace /dev/sdX with the right USB device; everything on it is erased.

14. Command line

Every step is also available in a terminal, in the order of the work. The commands apply each change at once (Review & Apply is only in the window).

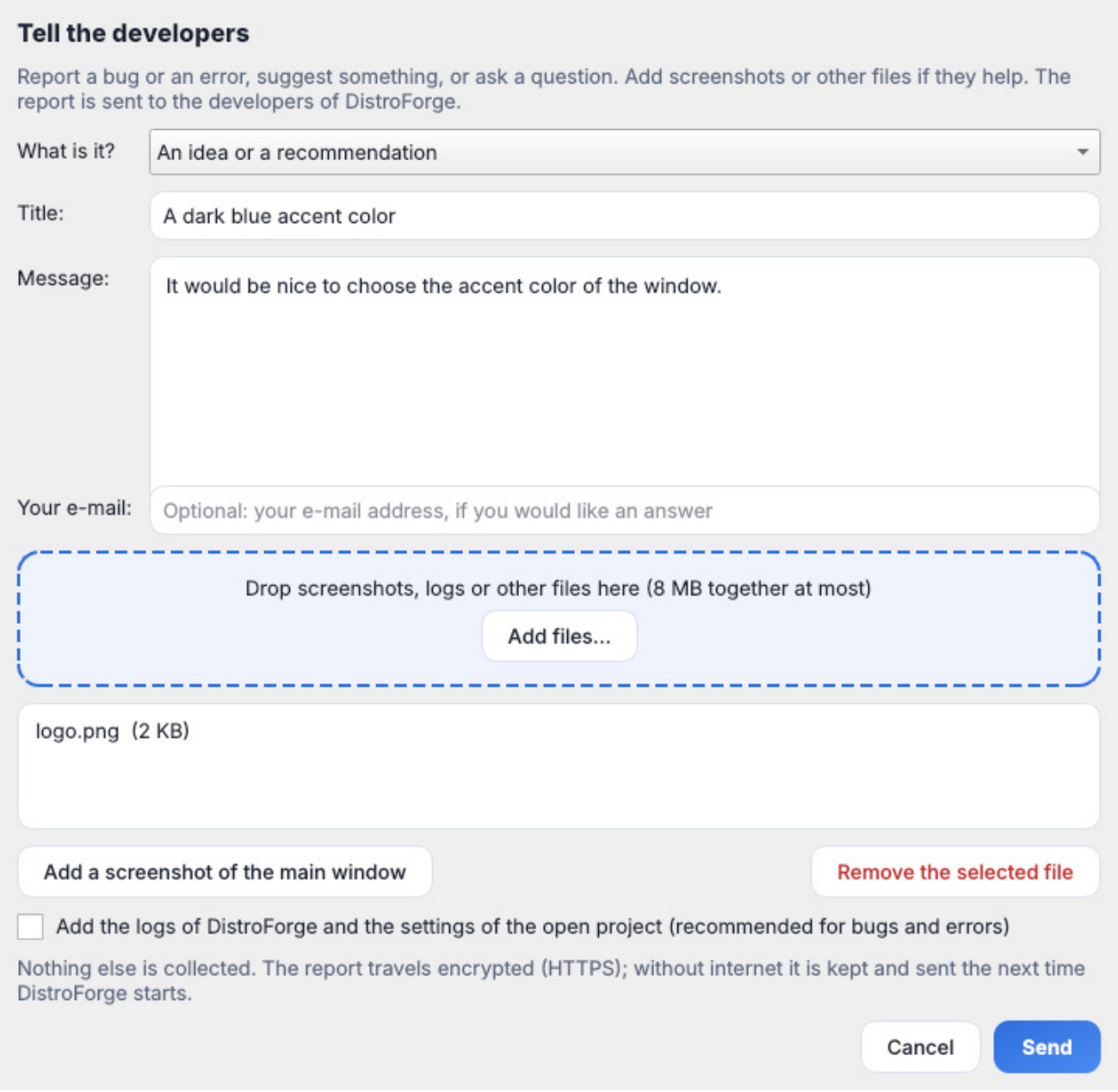
```
sudo distroforge new --iso debian-live.iso ~/mylinux
cd ~/mylinux
sudo distroforge brand identity name="My Linux" version=1.0
sudo distroforge apt install vlc gimp
sudo distroforge boot-loader list
sudo distroforge boot-loader set systemd-boot TIMEOUT=3
sudo distroforge boot-loader use systemd-boot
sudo distroforge check --fix
sudo distroforge build --target-size 700
sudo distroforge test
sudo distroforge clean --keep-iso
```

`distroforge --help` lists the commands by step; `distroforge COMMAND --help` explains one command.

15. Common problems and feedback

- **'grub needs /usr/sbin/grub-install in the image'** — not reported for live ISOs that install GRUB from the ISO pool during installation. If it shows, press Fix automatically.
- **Half-installed packages** — Fix automatically runs dpkg --configure -a and apt-get -f install.
- **Not enough disk space** — remove caches with `sudo distroforge clean`, or keep only the ISO of finished projects.
- **No internet** — installing packages and Flatpaks needs internet; all other edits work offline.
- **Log** — every message is in the log panel and in /tmp/distroforge/distroforge.log.

Send feedback (in the sidebar, in Settings, and in every error message) sends a bug report, an error, an idea or a question to the developers, with screenshots or other files and, if you agree, the logs. Without internet the report is kept and sent the next time DistroForge starts.

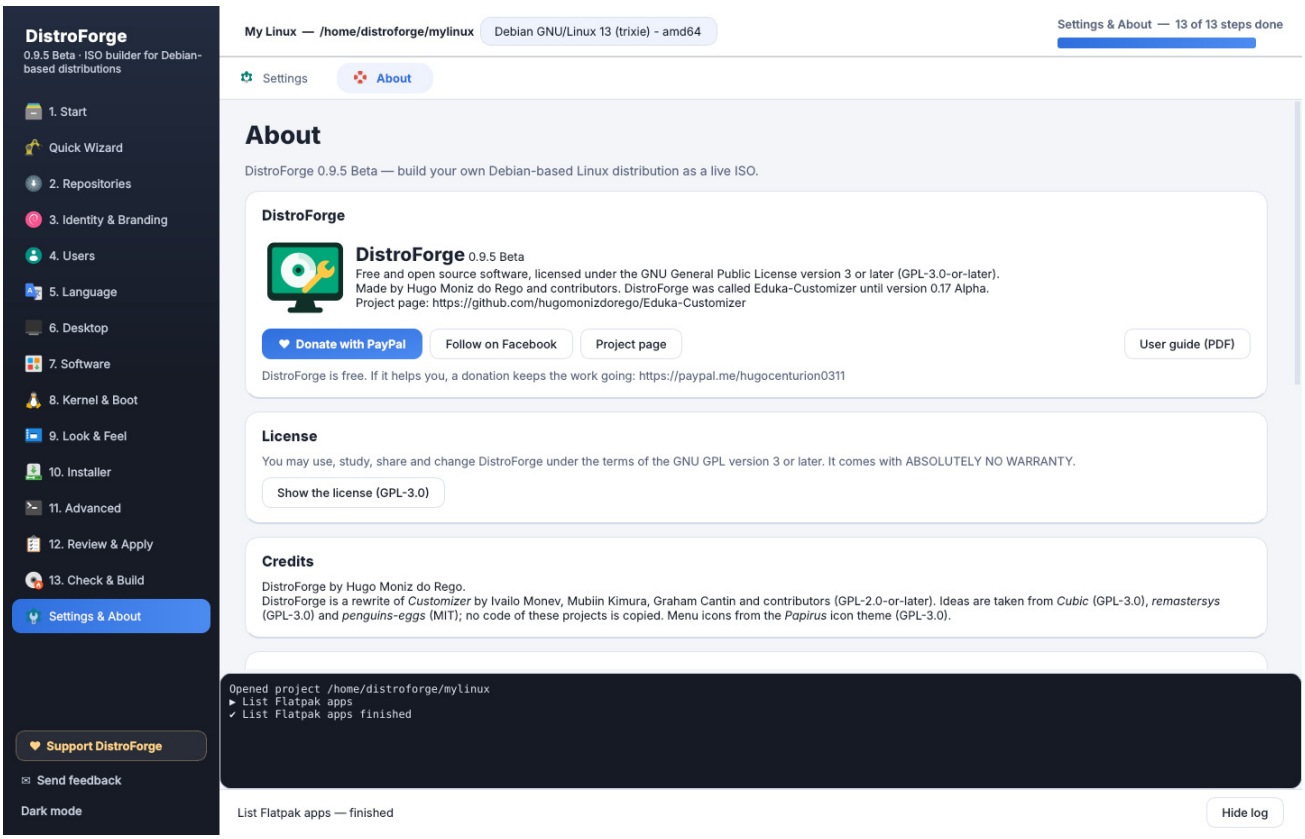


Send feedback

16. License, trademarks and support

DistroForge is free software under the GNU General Public License version 3 or later, WITHOUT ANY WARRANTY. The license of every component (Python, Qt, squashfs-tools, xorriso, GRUB, systemd-boot, rEFInd, Syslinux, Calamares, live-boot, Papirus, ...) is listed on the **About** tab of Settings & About. The packages in your ISO each keep their own license (/usr/share/doc/PACKAGE/copyright).

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About

Made by Hugo Moniz do Rego. If DistroForge helps you, a donation keeps the project going:

- PayPal: https://paypal.me/hugocenturion0311
- Facebook: https://facebook.com/hugomonizdorego
- Project: https://github.com/hugomonizdorego/Eduka-Customizer